

Lab 5 — Coverage

FIFO

Task 1

Covergroup Bins

The testbench defines one `covergroup` (`fifo_cov`) and one `cover` property directive. In total there are **13 covergroup bins + 1 cover directive**.

Coverpoint full (1 bin)

1. `went_full` — bin for `full == 1`

Coverpoint empty (1 bin)

2. `went_empty` — bin for `empty == 1`

Coverpoint `dut.count` (3 bins)

3. `empty_bin` — occupancy equals 0
4. `mid_bins` — occupancy in range `[1 : DEPTH-1]`
5. `full_bin` — occupancy equals `DEPTH`

Coverpoint `wr_en` (2 auto-bins)

6. `auto[0]` — `wr_en == 0`
7. `auto[1]` — `wr_en == 1`

Coverpoint `rd_en` (2 auto-bins)

8. `auto[0]` — `rd_en == 0`
9. `auto[1]` — `rd_en == 1`

Cross `wr_en` × `rd_en` (4 bins)

10. `write_only` — `wr_en=1, rd_en=0`
11. `read_only` — `wr_en=0, rd_en=1`
12. `both_off` — `wr_en=0, rd_en=0`
13. `illegal` — `wr_en=1, rd_en=1`

Cover Directive (+1)

- D1. `cover property (becomes_empty_after_read)` — covers the sequence where the FIFO transitions to empty after a read when `dut.count > 0`.

Bins Covered After Reset Only

The following **5 bins** are covered just from reset:

1. `went_empty` — after reset the FIFO is empty, so `empty == 1`
2. `empty_bin` — `dut.count == 0` after reset
3. `wr_en auto[0]` — `wr_en` stays 0
4. `rd_en auto[0]` — `rd_en` stays 0
5. `both_off` — cross bin where `wr_en=0`, `rd_en=0`

Task 2

Additionally Covered Bins

After enabling the provided stimulus, the following **5 additional bins** are now covered:

1. `write_only` — cross bin where `wr_en=1`, `rd_en=0`
2. `read_only` — cross bin where `wr_en=0`, `rd_en=1`
3. `wr_en auto[1]` — `wr_en == 1`
4. `rd_en auto[1]` — `rd_en == 1`
5. `mid_bins` — occupancy in range `[1 : DEPTH-1]`